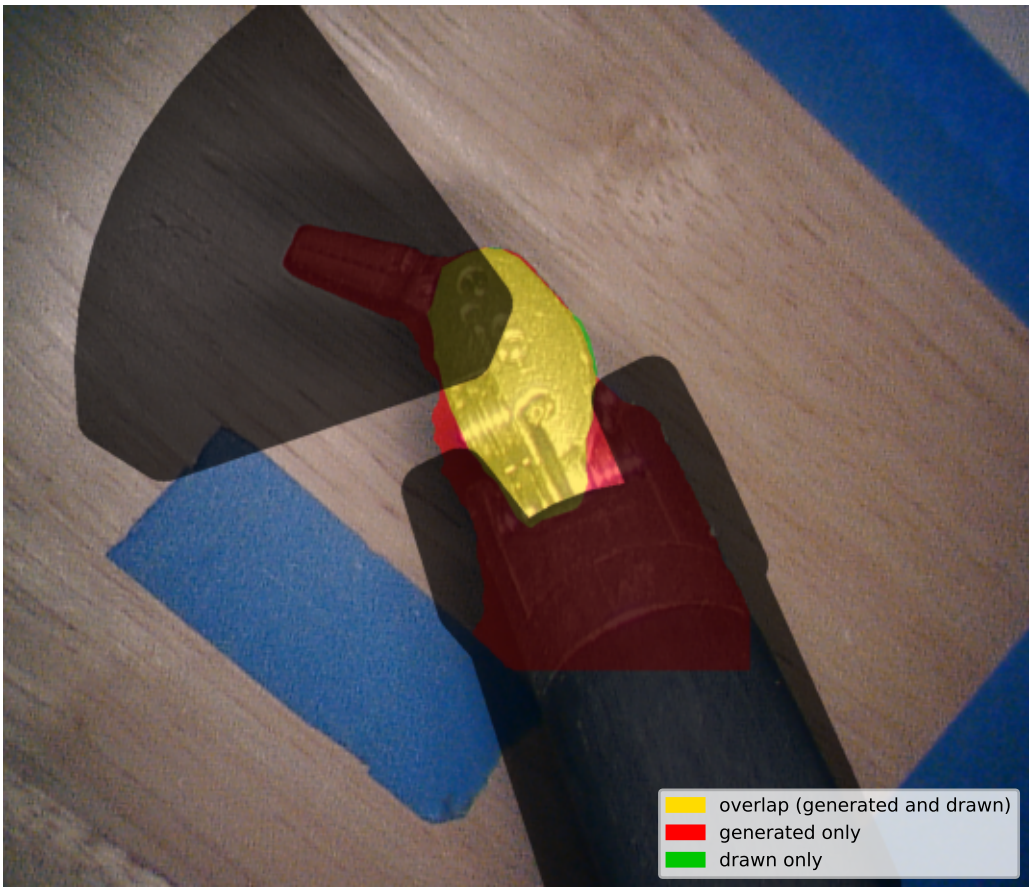
Generated masks vs hand segmentation

```
dataset:       /home/aseshad4/hand_eye_collection_scripts/gt_dataset
evaluated:     Wed Nov  4 16:21:11 2026
generated:     diff-render pipeline final-pass masks -- REBUILT (the run predates saving its masks)
               fit of 2026-11-01T16:32:34; same mask steps, last pass seeded from its final calibration (its pass-3
               estimate was not saved)
               /home/aseshad4/hand_eye_collection_scripts/gt_dataset/hand_eye_diff_render_comparison/masks
hand masks:    /home/aseshad4/hand_eye_collection_scripts/gt_dataset/hand_segmentation (clevis only)
frames:        [0, 2, 3, 5, 6, 8, 9, 11, 12, 14] (10 of the 15 training frames, evenly spaced)
model:         /home/aseshad4/TEST/joint.stl (clevis only)
IoU:           generated mask vs drawn mask, whole image; 'DR rgn' = only the pixels the diff-render
               loss uses (jaw / shaft / ring-prong zones excluded -- the generated mask covers them)
errors:        pose best fitting the generated mask vs pose best fitting the drawn mask (camera frame,
               at the pitch pin; gt_segmentation_eval.py's fit, 4 starts, DR region only);
               depth = camera z, only weakly observable from a silhouette
```

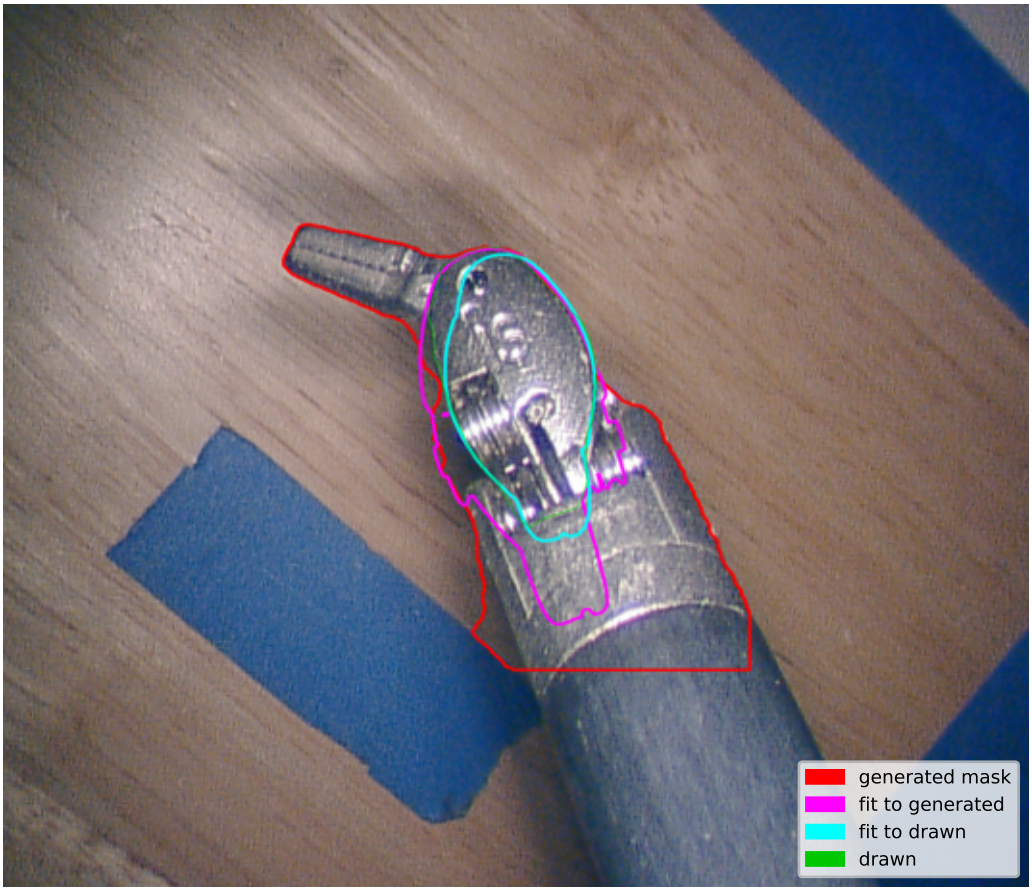
frame	IoU	IoU (DR rgn)	rot err [deg]	trans err [mm]	in-plane [mm]	depth [mm]
0	0.332	0.874	22.665	16.926	2.519	16.737
2	0.166	0.653	30.607	3.277	3.167	0.839
3	0.212	0.841	20.879	28.861	3.210	28.682
5	0.454	0.927	14.295	7.250	1.920	6.991
6	0.276	0.873	19.674	5.921	1.889	5.612
8	0.143	0.674	36.384	42.091	4.351	41.866
9	0.145	0.605	38.670	3.257	2.892	1.499
11	0.450	0.840	22.463	7.357	1.754	7.145
12	0.455	0.894	8.628	7.989	0.170	7.987
14	0.351	0.904	21.983	6.542	1.433	6.384
mean	0.298	0.809	23.625	12.947	2.331	12.374
median	0.304	0.857	22.223	7.304	2.220	7.068
std	0.122	0.112	8.820	12.184	1.099	12.503
max	0.455	0.927	38.670	42.091	4.351	41.866

frame 0 (right_rect-0000.png)   IoU 0.332 (DR region 0.874)
rot err 22.66 deg   trans err 16.93 mm (x -0.87, y +2.37, z -16.74 mm; in-plane 2.52)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)

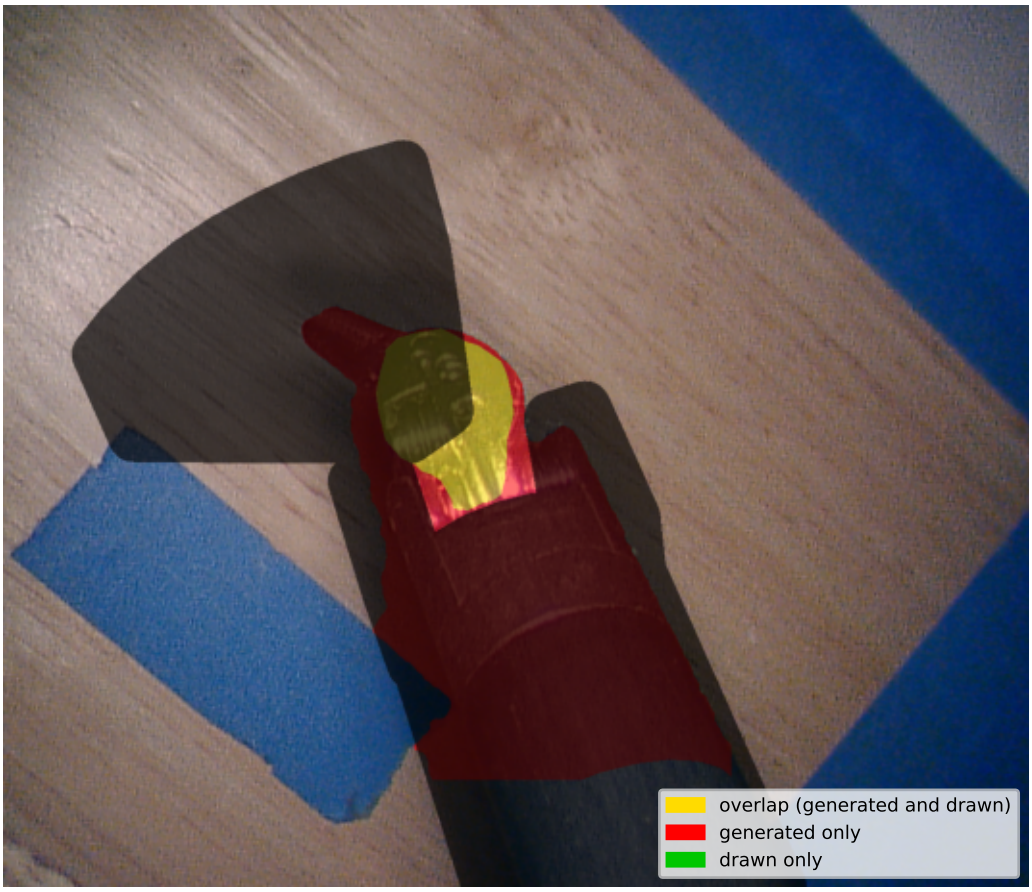


Errors: pose best fitting the generated mask vs pose best fitting the segmentation

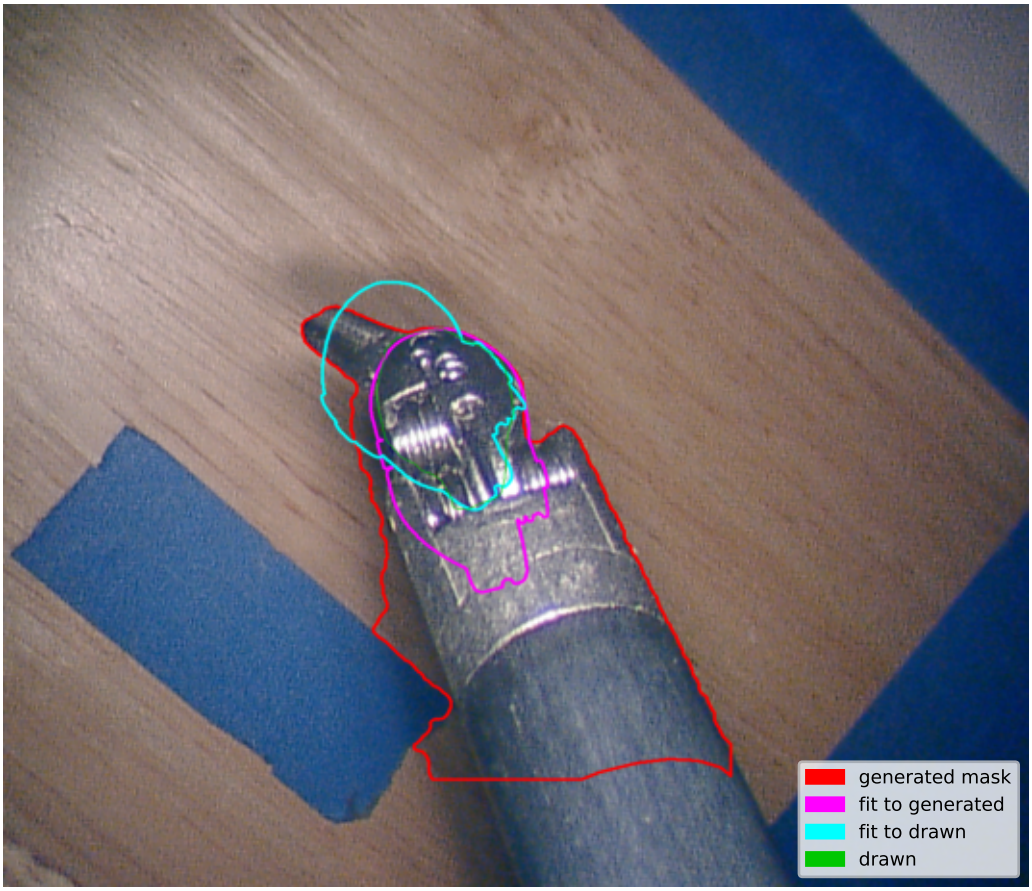


frame 2 (right_rect-0002.png)   IoU 0.166 (DR region 0.653)
rot err 30.61 deg   trans err 3.28 mm (x +1.33, y +2.87, z -0.84 mm; in-plane 3.17)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)

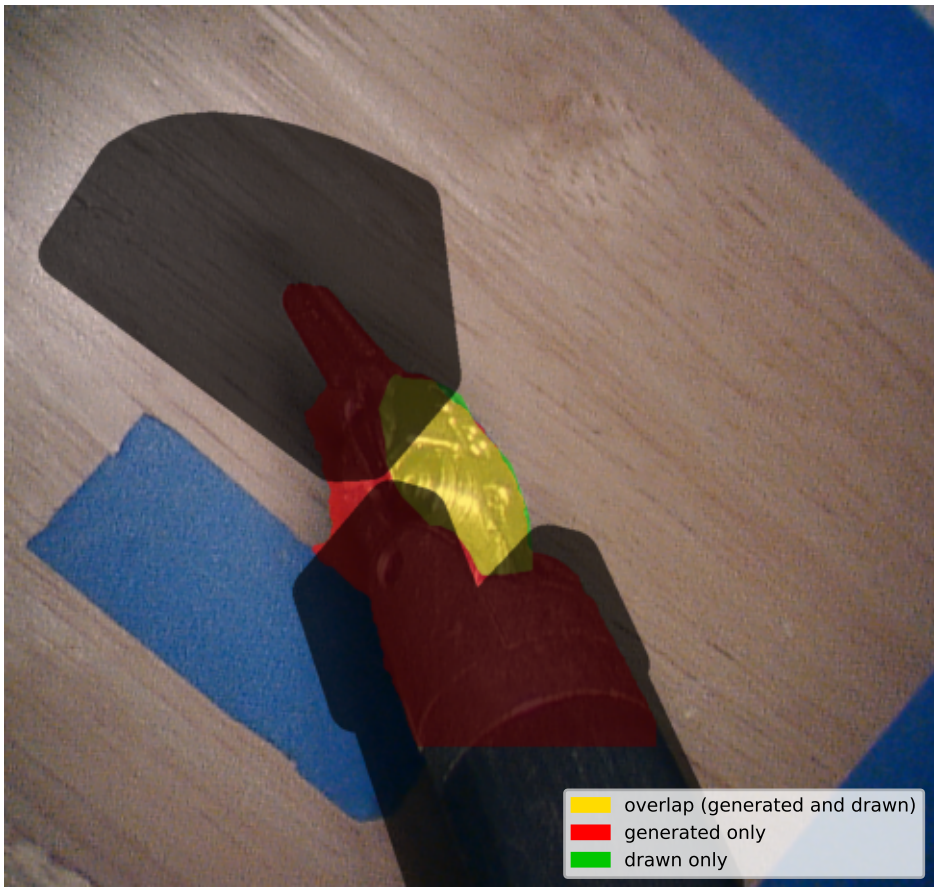


Errors: pose best fitting the generated mask vs pose best fitting the segmentation

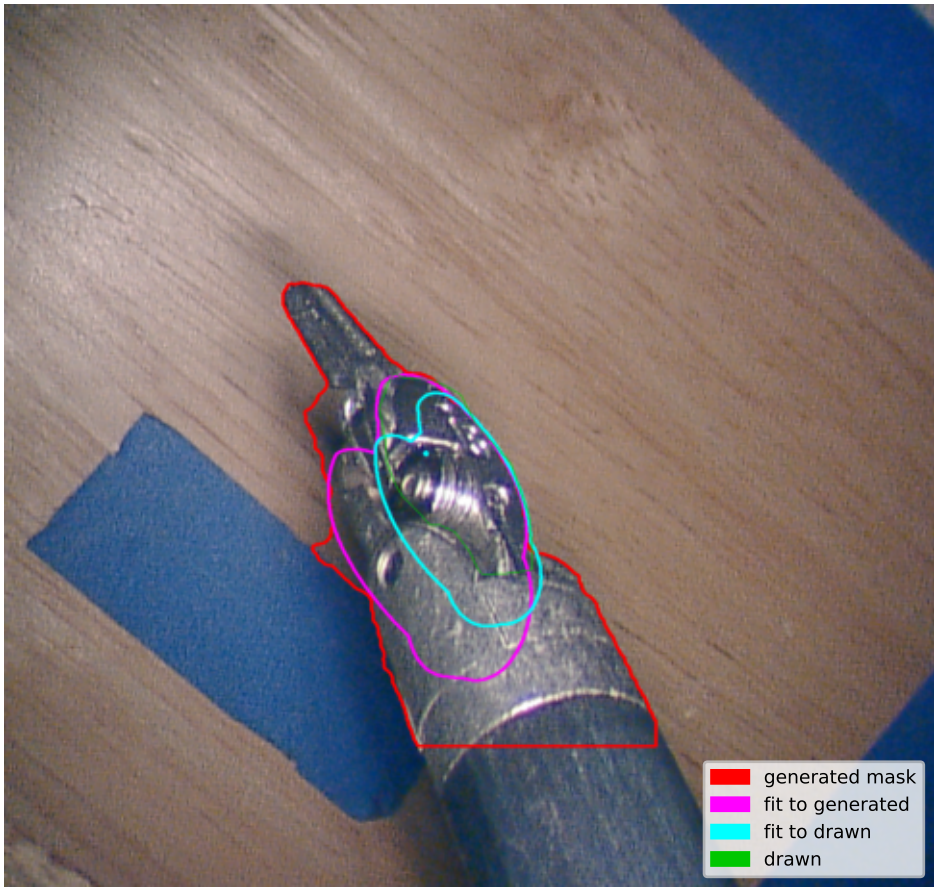


frame 3 (right_rect-0003.png)   IoU 0.212 (DR region 0.841)
rot err 20.88 deg   trans err 28.86 mm (x -3.15, y -0.60, z -28.68 mm; in-plane 3.21)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)

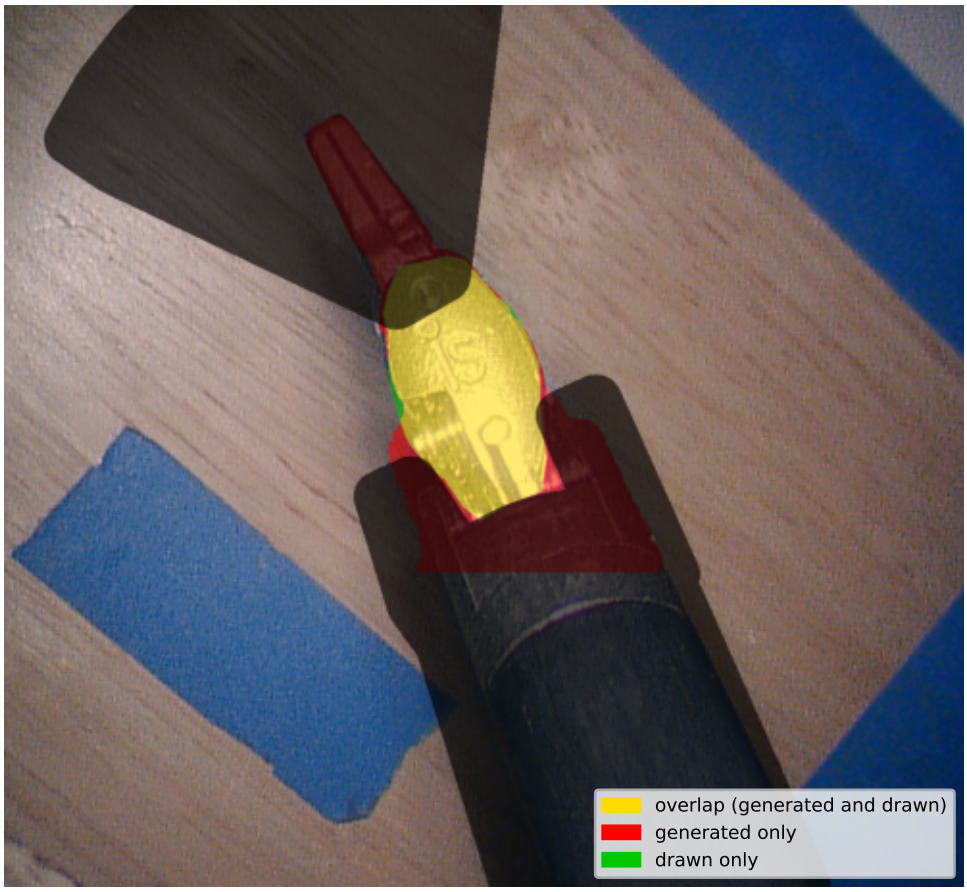


Errors: pose best fitting the generated mask vs pose best fitting the segmentation

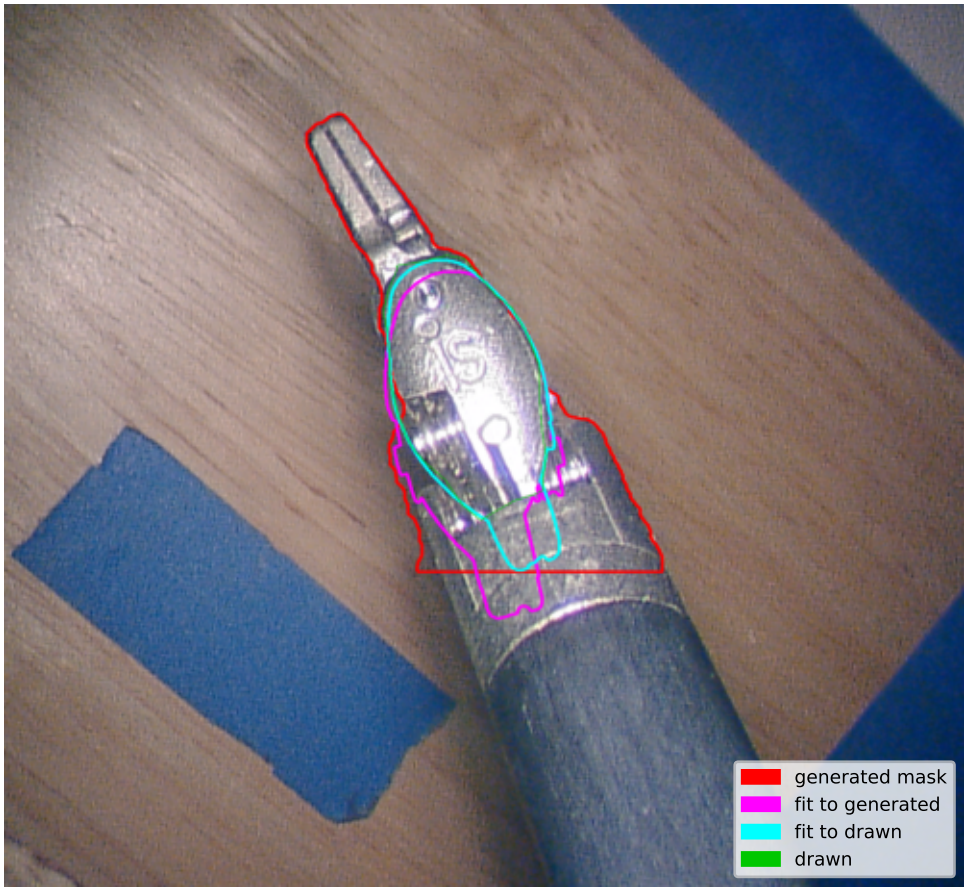


frame 5 (right_rect-0005.png)   IoU 0.454 (DR region 0.927)
rot err 14.29 deg   trans err 7.25 mm (x -1.26, y +1.45, z -6.99 mm; in-plane 1.92)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)

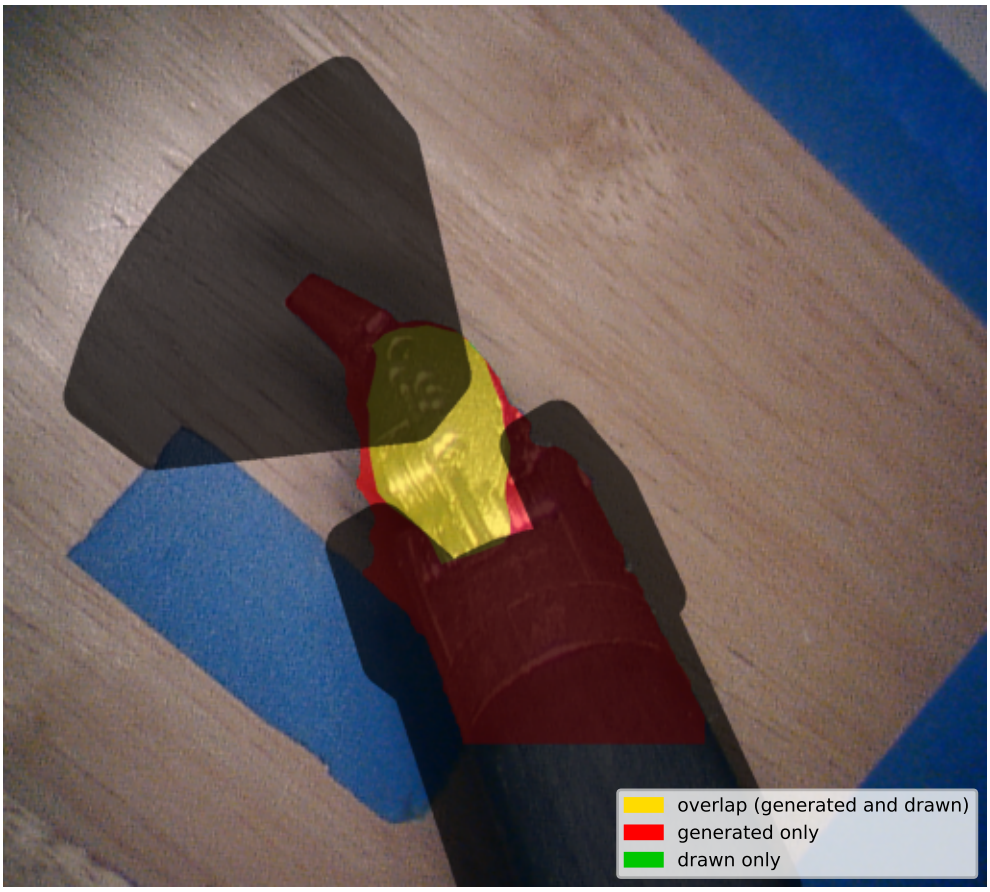


Errors: pose best fitting the generated mask vs pose best fitting the segmentation

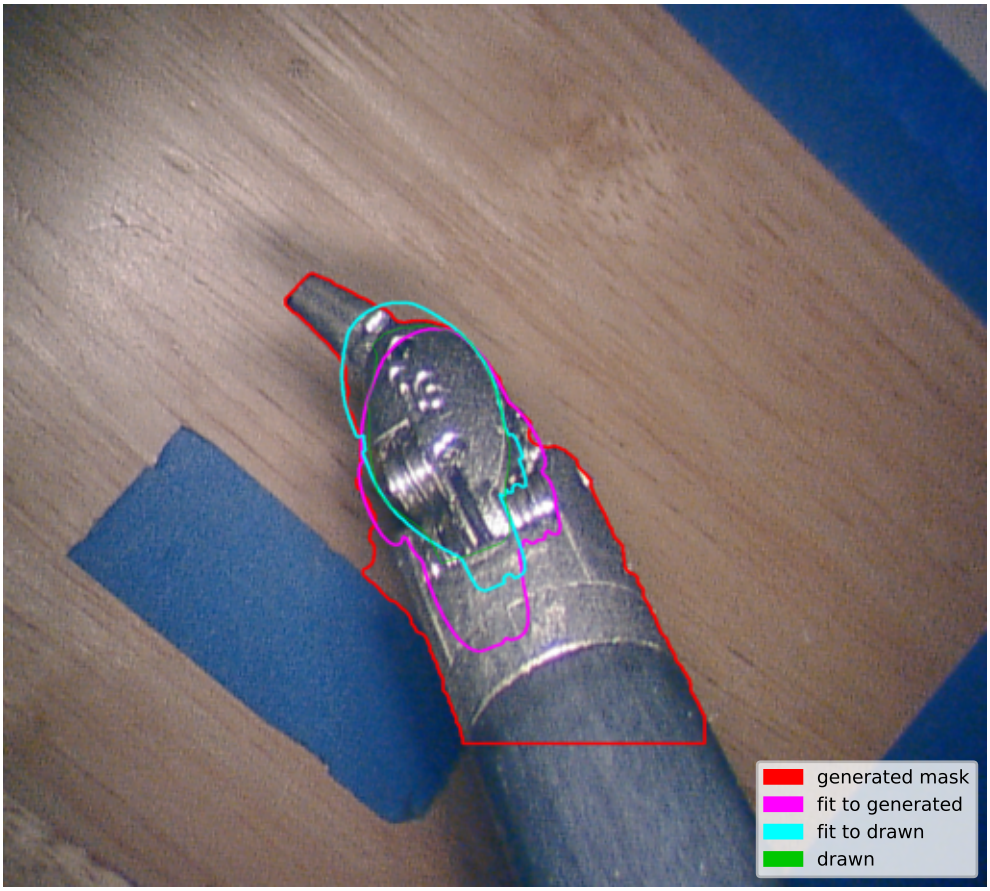


frame 6 (right_rect-0006.png)   IoU 0.276 (DR region 0.873)
rot err 19.67 deg   trans err 5.92 mm (x -0.35, y +1.86, z -5.61 mm; in-plane 1.89)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)

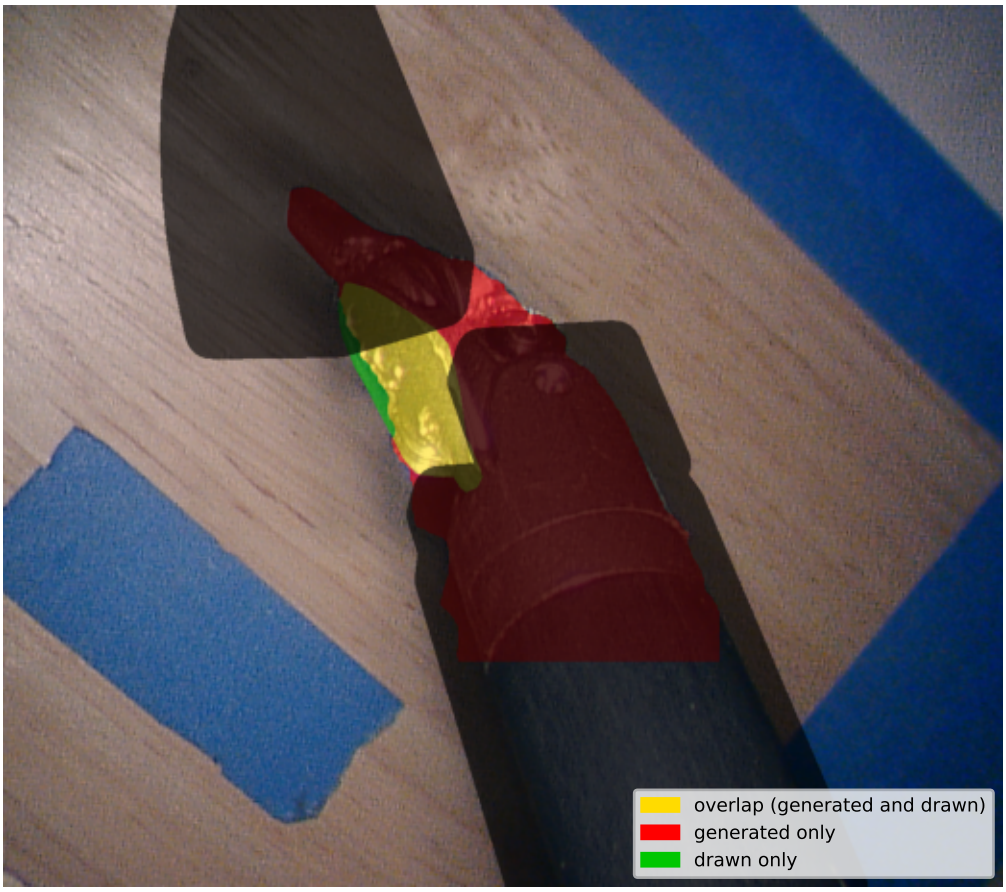


Errors: pose best fitting the generated mask vs pose best fitting the segmentation

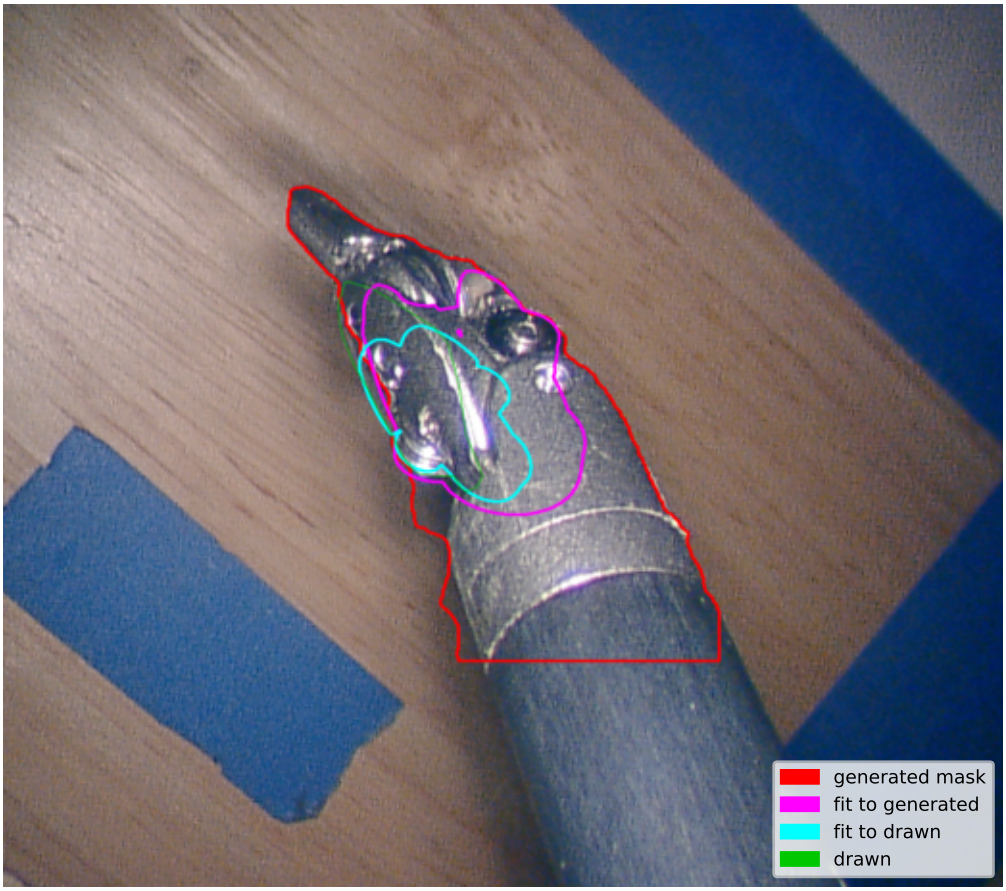


frame 8 (right_rect-0008.png)   IoU 0.143 (DR region 0.674)
rot err 36.38 deg   trans err 42.09 mm (x -4.33, y +0.42, z -41.87 mm; in-plane 4.35)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)

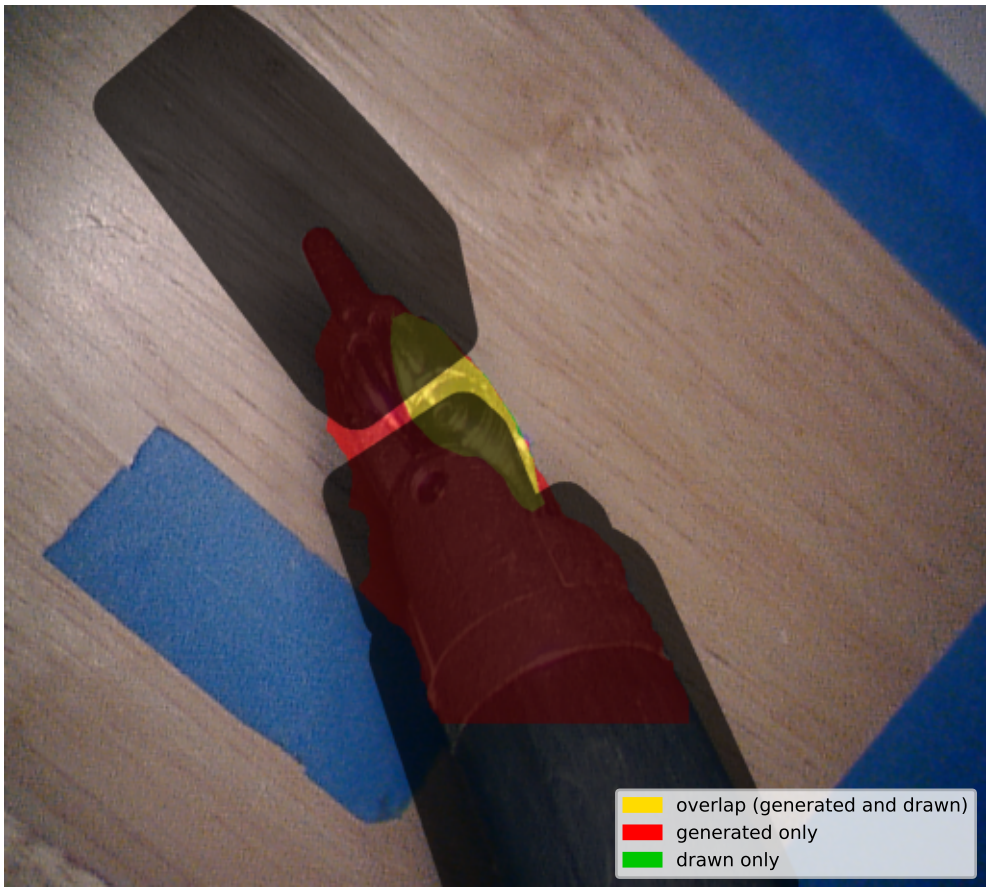


Errors: pose best fitting the generated mask vs pose best fitting the segmentation

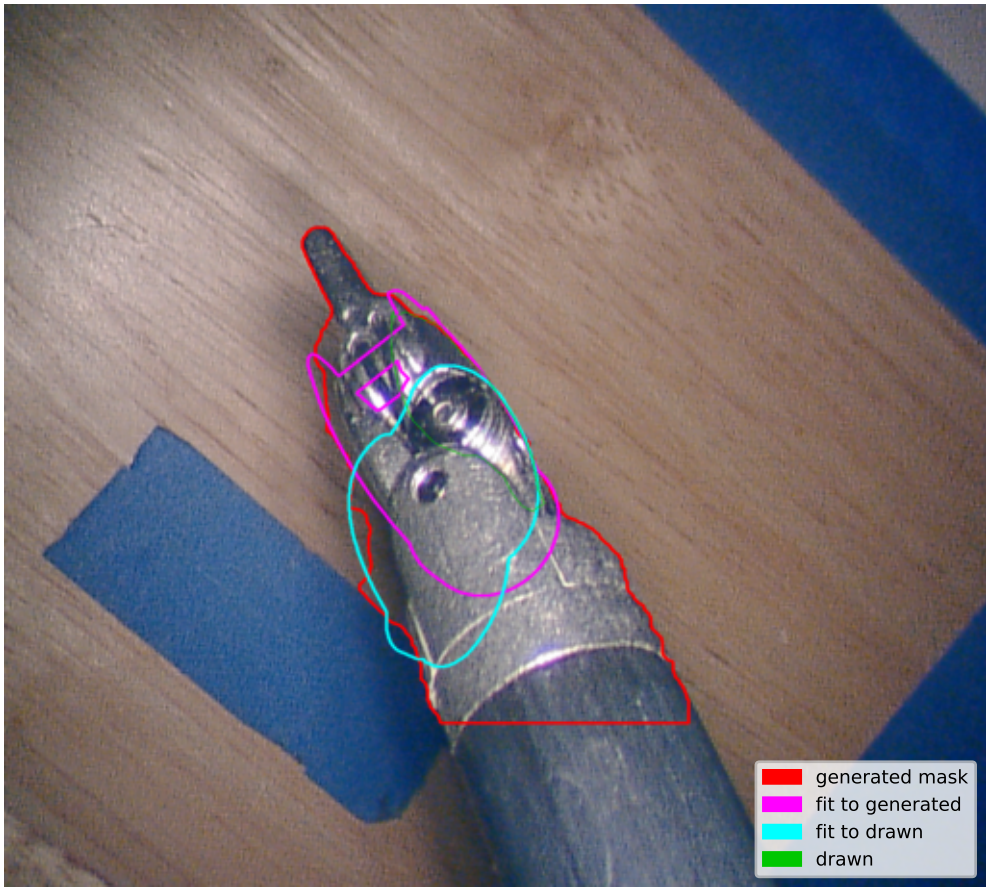


frame 9 (right_rect-0009.png)   IoU 0.145 (DR region 0.605)
rot err 38.67 deg   trans err 3.26 mm (x +1.43, y -2.51, z +1.50 mm; in-plane 2.89)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)

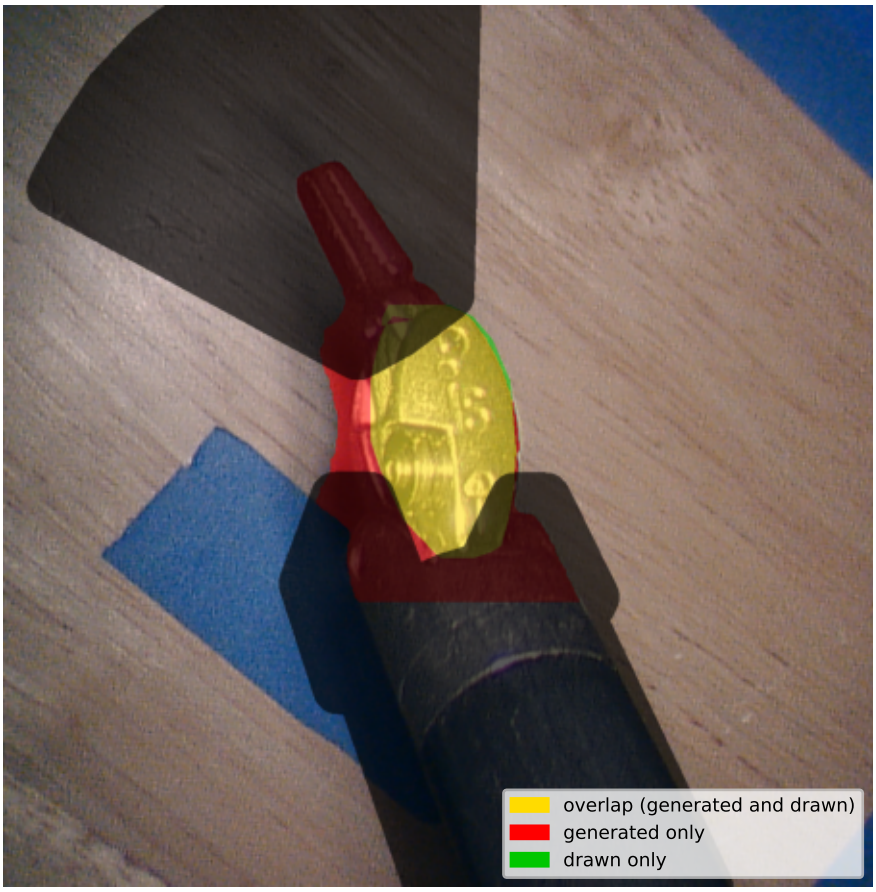


Errors: pose best fitting the generated mask vs pose best fitting the segmentation

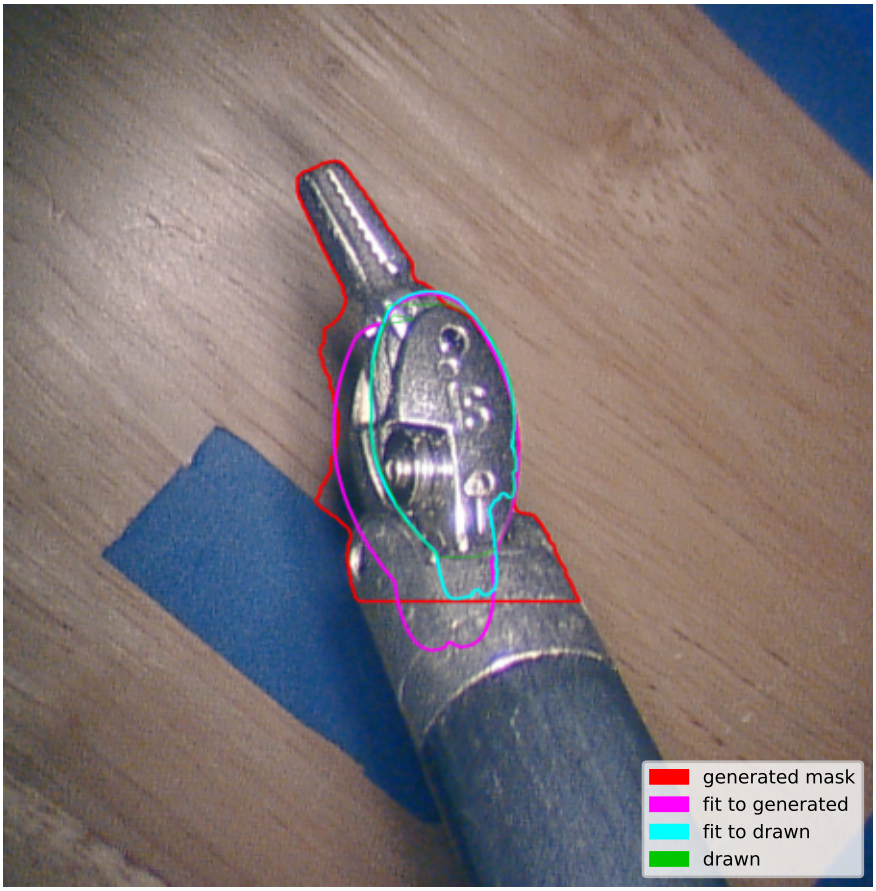


frame 11 (right_rect-0011.png) IoU 0.450 (DR region 0.840)
rot err 22.46 deg trans err 7.36 mm (x -0.91, y +1.50, z -7.15 mm; in-plane 1.75)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)

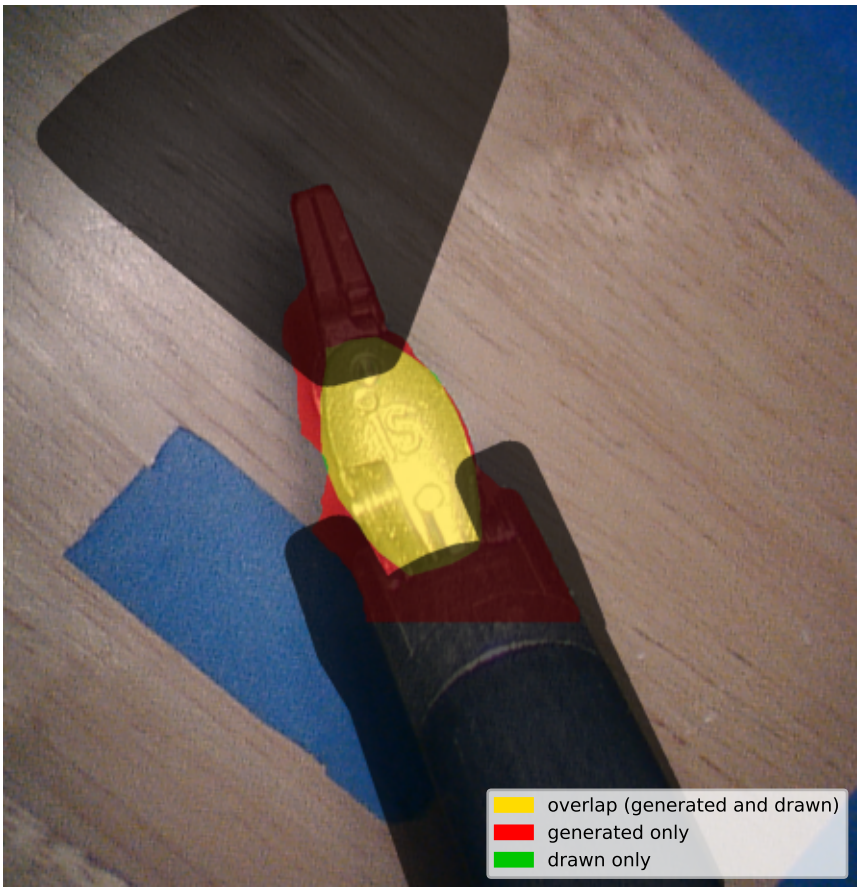


Errors: pose best fitting the generated mask vs pose best fitting the segmentation



frame 12 (right_rect-0012.png)   IoU 0.455 (DR region 0.894)
rot err 8.63 deg   trans err 7.99 mm (x +0.06, y -0.16, z -7.99 mm; in-plane 0.17)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)

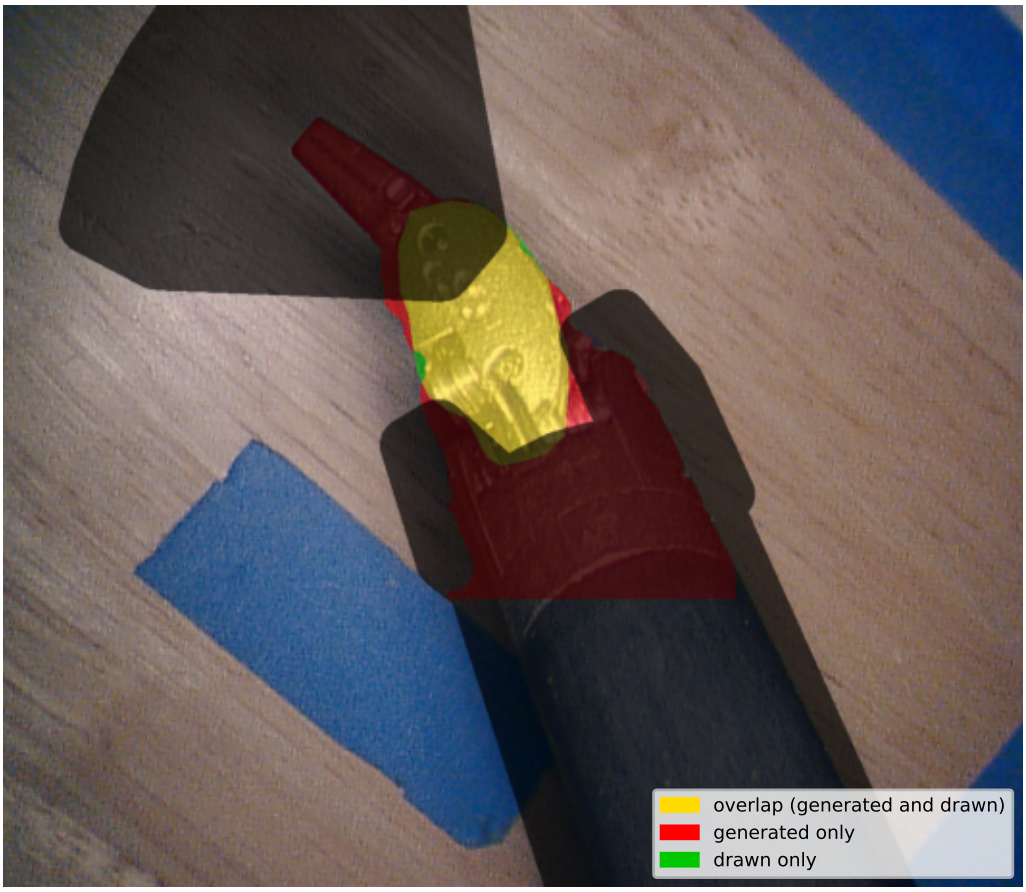


Errors: pose best fitting the generated mask vs pose best fitting the segmentation



frame 14 (right_rect-0014.png)   IoU 0.351 (DR region 0.904)
rot err 21.98 deg   trans err 6.54 mm (x +1.41, y +0.24, z -6.38 mm; in-plane 1.43)

IoU: generated mask over the segmentation (darkened = ignored by the DR loss)



Errors: pose best fitting the generated mask vs pose best fitting the segmentation

